

# Executive Summary: Customer Retention Analysis

By: Emmet Andrews for Github Example Project

## Data Context (via Kaggle / UCI)

The Online Retail II data set contains all the transactions occurring for a UK-based and registered, non-store online retail between 01/12/2009 and 09/12/2011. The company mainly sells unique all-occasion gift-ware. Many customers of the company are wholesalers.

## Objective

Analyze customer behavior using online retail transactional data (Online Retail II, 2009–2011) to understand retention patterns by cohort and identify opportunities to improve customer lifetime value.

## Key Findings

- **Steep early drop-off:** Month to month repeat customer retention falls to 21% of the original onboarding cohort after 1 month from first purchase. Meaning four out of five customers do not return after their first purchase the following month.
- **Flat long tail:** After Month 3, retention stabilizes at 15–20% of the original cohort, this is seen across all cohorts. A small loyal core continues to buy regularly.
- **Consistent patterns:** Retention curves are nearly identical across time. Newer cohorts (2010–2011) did not retain better than older ones.
- **Cohort size trend:** The largest cohort was Dec 2009 (~955 customers) and had a higher retention. Later cohorts were smaller (300–400 customers) and showed no improvement in loyalty between them.
- **Revenue growth driver:** Rising revenue and active users during 2010–2011 were powered by new acquisitions, not by stronger retention.

## Business Implications

- The current growth model is acquisition-led, but long-term value is undermined by poor retention.
- The critical window for retention is Month 1. Customers who return once are much more likely to become part of the loyal base.
- Marketing dollars may be better spent on onboarding, re-engagement, and loyalty programs rather than purely on acquisition.

## Recommendations

1. **Strengthen onboarding:** Welcome campaigns, personalized offers, and reminders within the first 30 days.
2. **Trigger re-engagement:** Target inactive customers after their first purchase with promotions or outreach.
3. **Segment loyal base:** Identify the ~15% who stay beyond Month 3, and explore cross-sell/up-sell opportunities.
4. **Track cohort metrics regularly:** Use dashboards to monitor whether new strategies shift the retention curves upward.

## Deliverables

- **SQL Pipeline (DuckDB):** Cleaned transactional data, created fact and retention tables.
- **Tableau Dashboard:** Interactive heatmap, retention curves, and KPI trends.
- **Insights Summary:** Cohort retention analysis highlighting Month 1 churn as the key driver of lifetime value.